

Database Vocabulary Card

Day 1 handout. The nine data-modeling terms to keep straight while you draft TMD Section 1 — with the TPM-relevant signal each one carries.

Use this as your reference while you model entities, keys, and indexes. You don't have to *implement* any of these — you have to name them clearly enough that an engineer can.

Term	What it means	TPM-relevant signal
Entity	A "thing" the system tracks (Dispatcher, Ticket, Reconcile)	The unit of the data model
Primary key	The unique identifier for an entity	Affects how entities are referenced
Foreign key	A reference from one entity to another	Models relationships
Index	A structure that speeds up specific reads at the cost of writes and storage	Tied directly to query patterns
Normalization	Splitting data into multiple entities to avoid duplication	Cleaner; sometimes slower
Denormalization	Duplicating data on purpose for read speed	Faster reads; harder consistency
Constraint	A rule the database enforces (NOT NULL, UNIQUE, CHECK)	Catches bugs before they ship
Cardinality	How many of one thing relate to how many of another (1:1, 1:N, N:N)	Governs the relationship shape
Schema	The set of definitions for entities + relationships + constraints	The artifact we ship today

The one discipline to remember: design the *queries* before the schema. If your data model can't be defended by the queries that drive it, it's wrong.